Professor Famularo Domenico

6 March 2026

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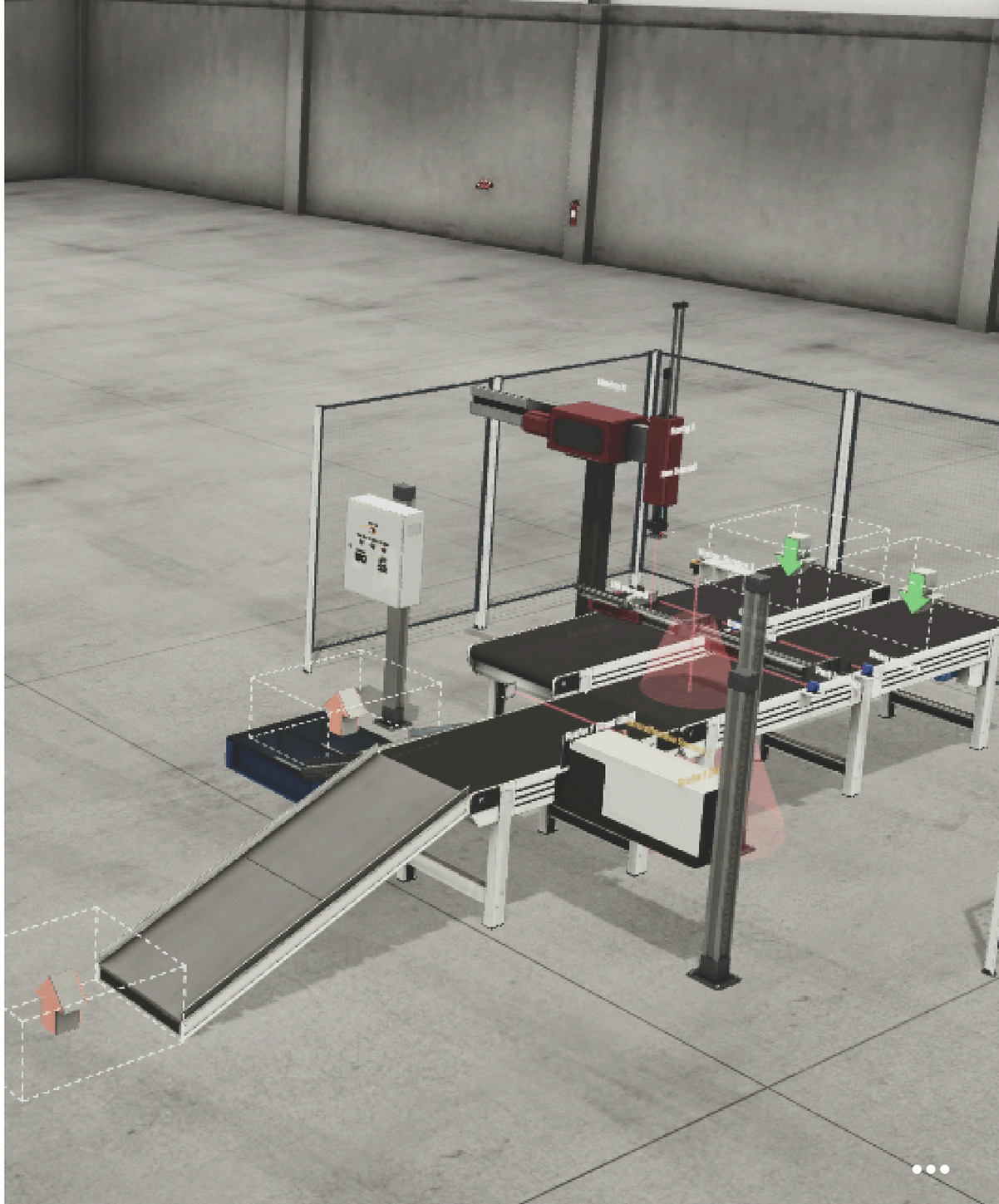
Industrial Automation Project



Industrial Automation Project

Table of contents

- Why create automated machines?
- Main goal of the project
- Project in factory io
- Structure of the code and connection
- Video of execution

Why create automated machines?

- **Increase Efficiency & Throughput**
- **Repeatability & Precision**
- **Safety & Ergonomics**

Automation is not just about making things faster, but about creating quality standards that a human could not maintain for hours at a time, while simultaneously increasing workplace safety.

Modern machines are no longer isolated islands. Thanks to the use of standard protocols like OPC UA, which I integrated into the project, the system can communicate instantly with company servers, providing real-time data on counting good parts and rejects.

Industrial Automation Project

The goal of this project is to design the **control logic (PLC)** for an intelligent automated assembly and sorting cell. In particular the sorting is **color based.** For this project two software were used: codesys and factory io

Project phases

Assembly phase

Sorting phase

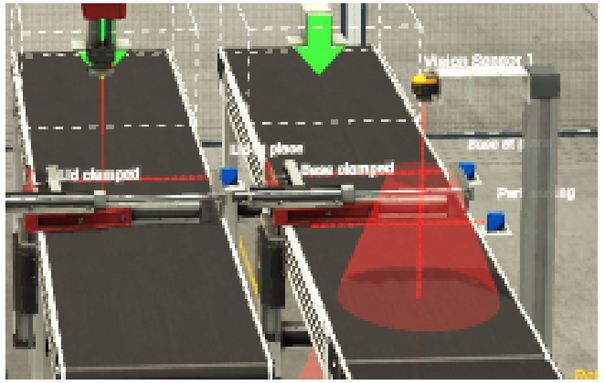
Main goal of the project

Project in

FACTORY I/O

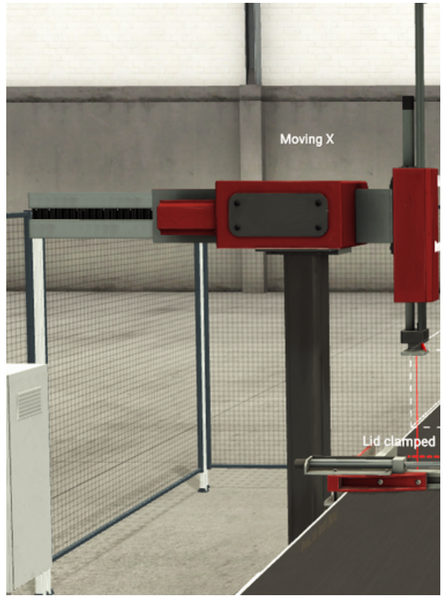
Electred Switchboard Controller

Used to control the execution, with the start buttons (necessary to start the assembly), the stop button which stops the tapes, and the reset button conceptualized as a sort of emergency stop.



Lids and Bases emitter and conveyor

They are used to slide the lids and bases.



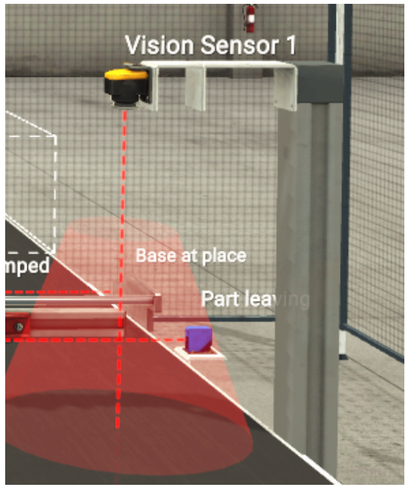
Pick and Place

Used to move the lid from the lids conveyor to the base conveyor, in order to complete the assembly.

Industrial Automation Project

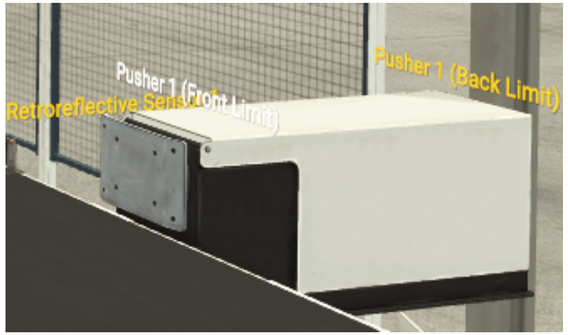
Project in

FACTORY I/O



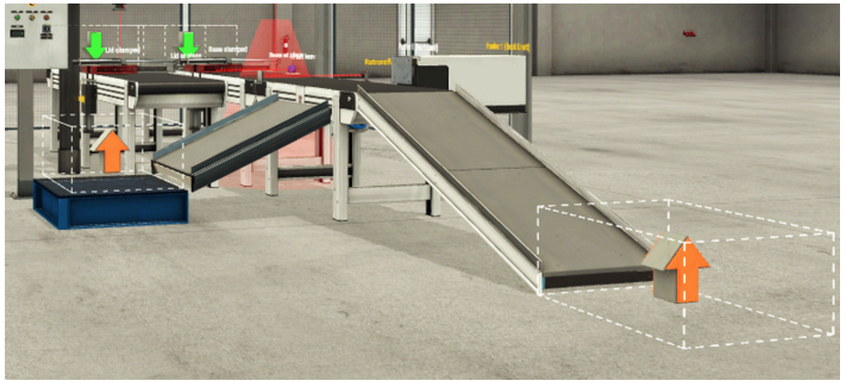
Vision Sensor

Vision sensor is configurated to give a bool output, in particular is true wen the lid of the pack is blue.



Pusher

When the packet passes through the retroflective sensor, and the color of the lid is blue the pusher pushes it.



Chute Conveyor

We need these for sorting.

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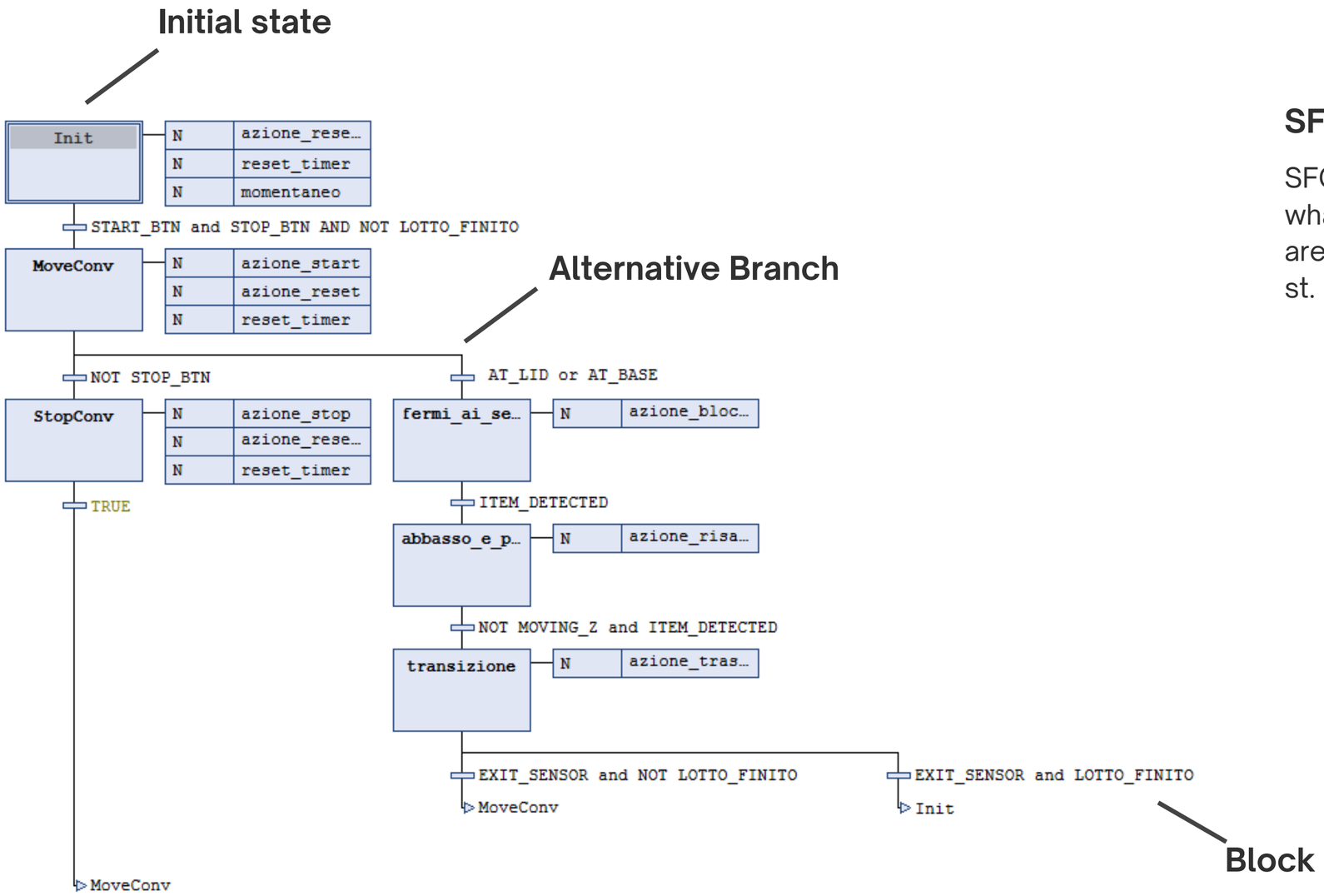
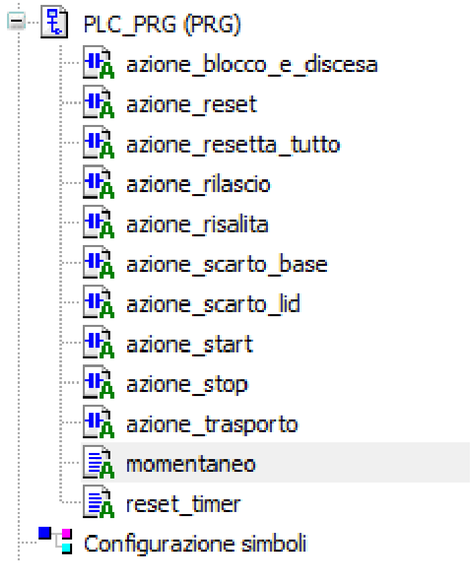
Code in codesys

CODESYS



SFC Structure

SFC was used to outline a sort of main of what is the actual program, in fact there are several actions written in ladder and in st.



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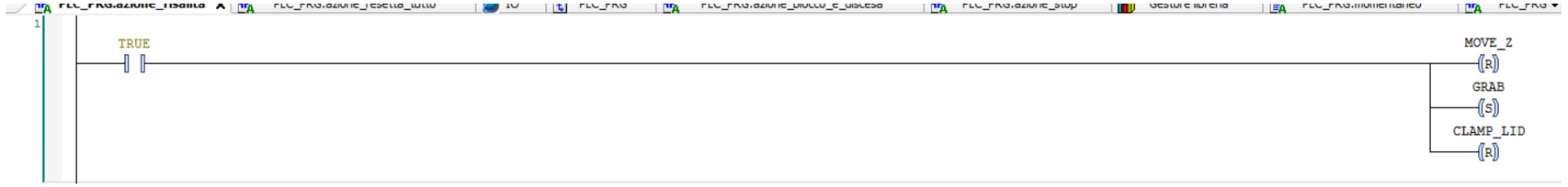
Code in codesys

Block and falling action

Rising action



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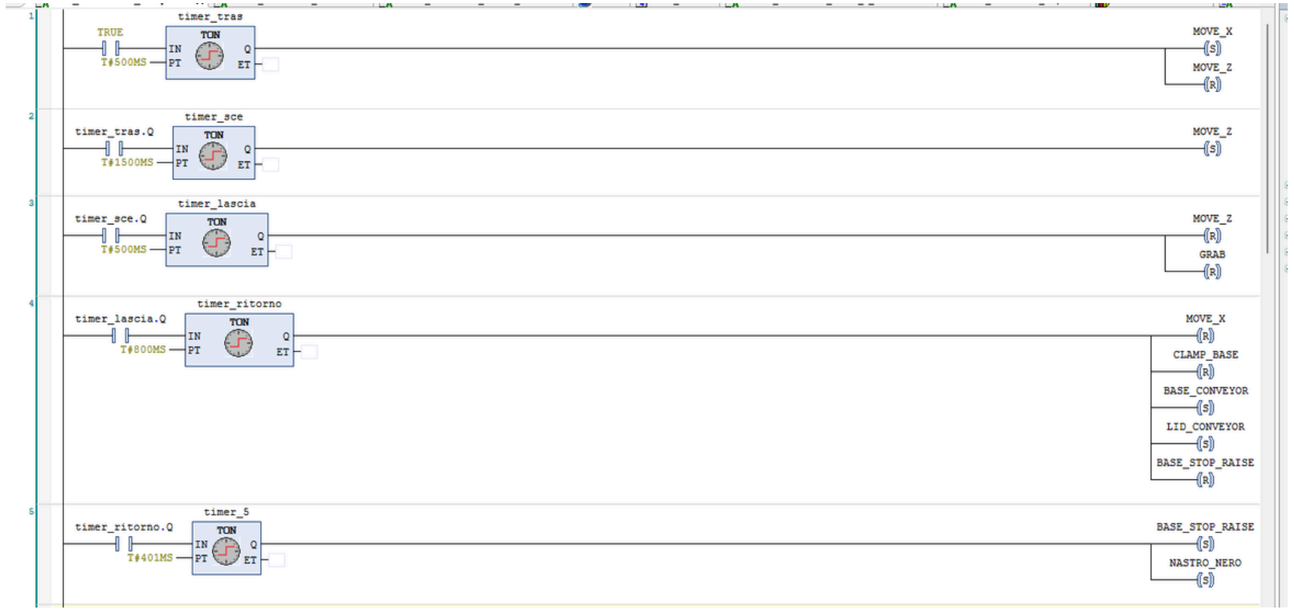
Code in codesys



Transport, counter and color sort action

PLC_PRG.reset_timer | PLC_PRG.azione_traspor

```
timer_tras(IN:=FALSE);
timer_sce(IN:=FALSE);
timer_lascia(IN:=FALSE);
timer_ritorno(IN:=FALSE);
timer_5(IN:=FALSE);
timer_sensor(IN:=FALSE);
TSL(IN:=FALSE);
TSB(IN:=FALSE);
TOF1(IN:=FALSE);
tim24(IN:=FALSE);
tim25(IN:=FALSE);
PRONTO_LID:=FALSE;
PRONTO_BASE:=FALSE;
timer_sbagliato_lid(IN:=FALSE);
tim_p(IN:=FALSE);
```

Timer reset

Industrial Automation Project

Video of execution